PROG7312 POE RESEARCH PART 1

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Introduction

The Smart-X IoT mesh takes in a constant stream of mixed telemetry from thousands of sensors spread across the network. As alerts pile up, a poorly designed dashboard can numb users to real problems and slow down troubleshooting. This report pinpoints five research-backed ways to boost dashboard engagement in the Smart-X data ingestion gateway, then takes a closer look at the strongest option – alert aggregation paired with contextual anomaly visualisation – to see how well it helps users catch and fix unusual data spikes and sensor dropouts.

Five Dashboard Engagement strategies

To tackle the risk of disengagement and missed anomalies in the Smart-X ingestion gateway's dashboard, five engagement strategies from current research are proposed:

1. Real-time visual telemetry feedback: Live-updating charts and colour-coded status indicators let operators watch sensor state changes as they happen, instead of relying on static or delayed reports, helping them spot abnormal readings faster (Bin Mofidul et al., 2022).
2. Proactive alert gamification: Framing configuration and troubleshooting tasks with progress or achievement markers has been associated with greater developer engagement, which in turn predicts job satisfaction (Stol, Schaarschmidt & Goldblit, 2022).
3. Simplified, chunked onboarding: Splitting first-time setup into small guided steps, rather than one dense manual, cuts the extraneous cognitive load on a learner's working memory (Sweller, 1988).
4. Progressive disclosure of advanced configuration: Hiding complex or rarely-used settings until asked for keeps the main interface simple. Though this doesn't always measurably lower cognitive load, it has been shown to improve users' perceived sense of learning and understanding of a system (Anik & Bunt, 2026).
5. Alert aggregation with contextual anomaly visualisation: Grouping related alerts and showing how they connect, instead of presenting each alone, tackles "Disconnected and Overloaded Dashboards," one of four main causes of alert fatigue in the SOC literature, and shows promise for faster, more accurate diagnoses (Tariq et al., 2025; Eckhoff et al., 2025).

Alert Aggregation with Contextual Anomaly Visualisation

Grouping related alerts instead of showing each one alone tackles a named root cause of alert fatigue head-on. Tariq et al. (2025) name "Disconnected and Overloaded Dashboards" as one of the four main causes of alert fatigue in security operations centres, alongside industry survey findings that over half of monitoring teams feel overwhelmed by alert volume, with a large share of analyst time spent triaging low-value notifications. When an operator faces a constant, unstructured stream of individual notifications, each one demands its own decision,

and the cognitive burden builds with every added alert. Grouping related alerts into a single event collapses several decisions into one, cutting total attentional load without losing any underlying information. For Smart-X, where thousands of microcontrollers constantly publish multi-typed telemetry, an ungrouped alert stream would hit the same scaling problem: as the mesh grows, so does alert volume, and with it the risk that operators go numb to genuinely important signals.

Beyond cutting volume, grouping alerts by shared properties preserves the links between them, letting an operator see cause rather than disconnected symptoms. Eckhoff et al. (2025) describe how alerts sharing properties, such as a device identifier or timestamp proximity, can be linked into graph-based groups that capture a single underlying event rather than a list of unrelated triggers. They report that 62% of all alerts in a typical security operations centre are ignored, underscoring how easily isolated, uncontextualized alerts get lost in high-volume monitoring environments. Applied to Smart-X, a single sensor disconnect could plausibly throw off several unrelated-looking alerts, for instance a power dropout that also produces a soil-moisture reading gap and a valve-state anomaly. Reviewed alone, these look like three separate problems; grouped by device and time-window, they turn out to be one underlying event. This directly supports spotting and fixing anomalous data spikes and sensor disconnects, since it turns several ambiguous alerts into one diagnosable incident instead of making an operator manually correlate them.

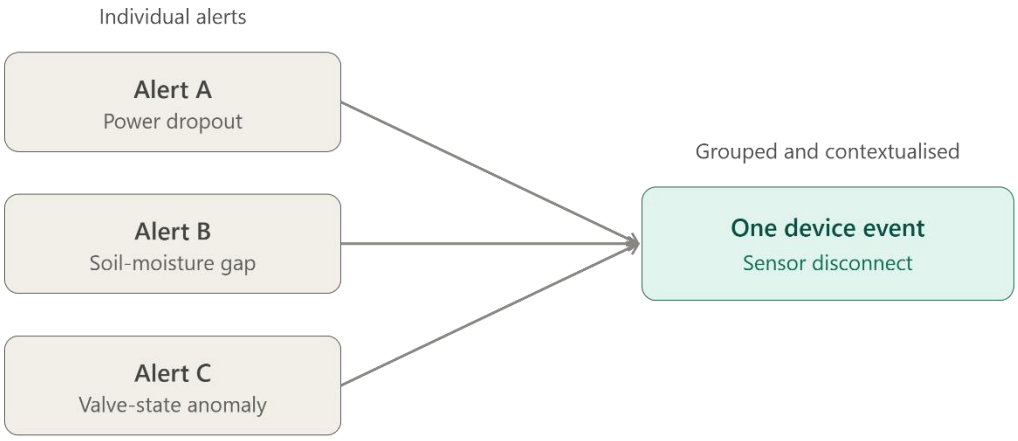


Figure 1: Three isolated alerts from a single sensor disconnect consolidated into one grouped, contextualised alert (adapted from the alert grouping concept in Eckhoff et al., 2025)

Both supporting sources come from cybersecurity monitoring rather than IoT, but this is no forced analogy. Tariq et al. (2025) explicitly note that alert fatigue is not exclusive to security operations centres, extending to other high-volume monitoring domains that demand constant operator attention, including energy control rooms and factory automation systems, both of which closely resemble the Smart-X environment. The underlying mechanism behind fatigue, a constant, high-volume, high-noise stream overwhelming a human's limited attention, is the same whether the alerts come from intrusion detection tools or IoT sensor telemetry. Because alert aggregation and contextual visualisation address that general mechanism rather than anything specific to cybersecurity, the strategy applies directly to Smart-X's dashboard design. This supports its selection as the most defensible of the five engagement strategies,

since it rests on research the original authors themselves identify as reaching beyond their own domain.

Conclusion

Poorly designed dashboards risk desensitising operators to the exact anomalies a monitoring system exists to catch. Of the five engagement strategies identified, alert aggregation with contextual anomaly visualisation offers the strongest, most directly evidenced response to this risk, converting scattered, ambiguous alerts into single diagnosable events and directly supporting faster identification of sensor disconnects and data spikes. While the strategy grew out of cybersecurity monitoring research, its underlying mechanism, human attention overwhelmed by high-volume alert streams, applies equally to Smart-X. This strategy should inform the design of the dashboard implemented in the ingestion and validation gateway, prioritising grouped, contextualised alerts over isolated notifications.

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